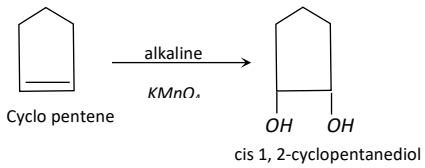
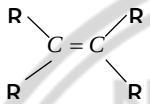
ALKENE

41. (c)



42. (a) We know that greater the number of alkyl groups attached to double bonded carbon atoms, more stable is the alkene. Therefore most stable is $R_2C = CR_2$



43. (c) $CH_2 = CH_2 + 2[O] \xrightarrow{KMnO_4} \underset{\text{Formaldehy de}}{HCHO} + HCHO$

44. (c) Paraffins are non-polar compounds. The intermolecular forces are weak Vander Waal's forces. As the molecular mass increases Vander Waal's forces increases. Hence boiling point increases.

45. (a) $\underset{Br}{\underset{|}{CH_2}} - \underset{Br}{\underset{|}{CH_2}} + Zn \rightarrow ZnBr_2 + CH_2 = CH_2$

46. (c) $CH_3CH_2I + KOH(alc) \rightarrow CH_2 = CH_2 + KI + H_2O$

$$\underset{\text{Alk.Pink}}{CH_2 = CH_2 + KMnO_4} \xrightarrow{H_2O} \underset{\text{Colourless}}{\underset{OH}{\underset{|}{CH_2}} - \underset{OH}{\underset{|}{CH_2}}} + MnO_2$$

47. (d) $CH_2 = CH - Br \xrightarrow{HBr} CH_3 - CH\begin{matrix} Br \\ Br \end{matrix}$

According to Markownikoff's rule *H* atom or positive part goes to that carbon atom which is more hydrogenated.

48. (d) $CH_3 - CH_2 - Br + \underset{(alc)}{KOH} \rightarrow \underset{\text{Ethene}}{CH_2 = CH_2} + KBr + H_2O$

49. (a) $CH_3-CH_2-OH \xrightarrow[\text{Dehydratio n}]{\text{Conc. } H_2SO_4} CH_2=CH_2+H_2O$

50. (c) Electrophillic addition reactions are shown by alkenes or alkynes in these reactions attacking species is electrophiles

$$CH_3-CH=CH_2+H^+ \xrightarrow{\text{Slow}} \underset{2^o \text{ carbonium ion}}{CH_3-\overset{+}{C}H-CH_3}$$

$$CH_3-\overset{+}{C}H-CH_3+Br^- \xrightarrow{\text{Fast}} \underset{\text{2-Bromo propane}}{CH_3-\underset{\underset{Br}{|}}{CH}-CH_3}$$

51. (a) $CH_3CH_2CH_2Cl+KOH \rightarrow CH_3-CH=CH_2+KCl+H_2O$

$$CH_3-\underset{\underset{Cl}{|}}{CH}-CH_3+KOH \rightarrow CH_3-CH=CH_2+KCl+H_2O$$

52. (d) $\underset{\text{Ethene}}{CH_2=CH_2} \xrightarrow{\text{Cold.alk.}KMnO_4} \underset{\text{Ethylene glycol}}{\underset{\underset{OH}{|}}{CH_2}-\underset{\underset{OH}{|}}{CH_2}}$

53. (c) Ethane and Methane does not decolourise $KMnO_4$ and does not react with ammonical cuprous chloride. Acetylene decolourise $KMnO_4$ solution and also gives red ppt. with ammonical cuprous chloride. On the other hand ethene decolourize $KMnO_4$ solution but does not react with ammonical cuprous chloride.

54. (a) $CH_2=CH_2+\overset{-}{H}O\overset{+}{C}l \rightarrow \underset{\text{1–Chloro-2-hydroxy ethane}}{\overset{2}{\underset{\underset{OH}{|}}{CH_2}}-\overset{1}{\underset{\underset{Cl}{|}}{CH_2}}}$

55. (b) $\underset{\text{(ethylene)}}{n(CH_2=CH_2)} \xrightarrow[\text{high pressure}]{400^oC} \underset{\text{(polyethylene)}}{(-CH_2-CH_2-)_n}$

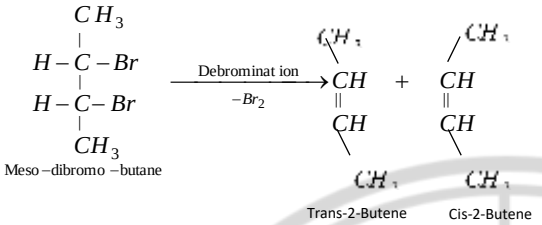
56. (c) CH_3–CH=CH–CH_2–CH_2–OH

- Decolourises bromine means presence of C=C (unsaturation).

- Gives white fumes of HCl with PCl_5 indicates presence of –OH group (alcohol).

Only option (c) has both C=C and –OH group.

57. (d)



Trans-2-butene is more stable than its *cis* isomer.

58. (a) $ClCH_2-CH_2-CH_2-CH_3 \xrightarrow{alc.KOH} \underset{1-butene}{CH_2=CH-CH_2-CH_3}$

59. (d) $CH_3CH_2CH=CHCH_3 \xrightarrow[Zn/H_2O]{O_3} \underset{Propanal}{CH_3CH_2CHO} + \underset{Ethanal}{CHOCH_3}$

60 (b) 1.35 Å

In ethylene (C_2H_4), carbon atoms are connected by a double bond, which is shorter than a single bond (1.54 Å).